

METODOLOGÍA CIENTÍFICA 1 – MÉTODOS CUANTITATIVOS

Medidas de Resumen:

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n} \quad s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1} \quad CV = \frac{s}{\bar{x}} \times 100\%$$

$$Md = l_i + \frac{0.5 - F_i}{F_{i+1} - F_i} \times (l_{i+1} - l_i) = l_i + \frac{0.5 - F_i}{f_i} \times (l_{i+1} - l_i)$$

Donde:

l_i	Límite inferior del intervalo mediano	F_{i+1}	Frecuencia relativa acumulada del intervalo mediano
l_{i+1}	Límite superior del intervalo mediano	f_i	Frecuencia relativa del intervalo mediano
F_i	Frecuencia relativa acumulada del intervalo anterior	A_i	Amplitud del intervalo mediano

Probabilidad – Teorema de Bayes

$$P(A|B) = \frac{P(B|A) \times P(A)}{P(B|A) \times P(A) + P(B|A^c) \times P(A^c)}$$

Distribuciones

$$X \sim \text{Binom}(n, p) \rightarrow P(X = x) = C_x^n \times p^x \times (1 - p)^{n-x} = \frac{n!}{x! \times (n - x)!} \times p^x \times (1 - p)^{n-x}$$

$$X \sim \text{Poisson}(\lambda) \rightarrow P(X = x) = \frac{e^{-\lambda} \times \lambda^x}{x!}$$

$$X \sim N(\mu, \sigma); Z = \frac{X - \mu}{\sigma} \rightarrow Z \sim N(0, 1)$$

Procedimientos Diagnósticos

$$VPP = \frac{s \times \text{prevalencia}}{s \times \text{prevalencia} + \alpha \times (1 - \text{prevalencia})} \quad VPN = \frac{e \times (1 - \text{prevalencia})}{e \times (1 - \text{prevalencia}) + \beta \times \text{prevalencia}}$$

$$\text{Odds} = \frac{p}{1 - p} \quad p = \frac{\text{Odds}}{1 + \text{Odds}} \quad LR^+ = \frac{s}{1 - e} \quad LR^- = \frac{1 - s}{e}$$

Inferencia

$$IC_{1-\alpha} = \bar{x} \pm z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}} \quad IC_{1-\alpha} = \bar{x} \pm t_{\alpha/2, n-1} \times \frac{s}{\sqrt{n}} \quad IC_{1-\alpha} = \hat{p} \pm z_{\alpha/2} \times \sqrt{\frac{\hat{p} \times (1 - \hat{p})}{n}}$$

$$n = \frac{(z_{\alpha/2})^2 \times \sigma^2}{I^2} \quad n = \frac{(z_{\alpha/2})^2 \times \hat{p} \times (1 - \hat{p})}{I^2}$$

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Riesgo

$$\widehat{OR} = \frac{a \times d}{b \times c} \quad IC_{1-\alpha} = e^{\ln \widehat{OR} \pm z_{\alpha/2} \times EE(\ln \widehat{OR})} \quad EE(\ln \widehat{OR}) = \sqrt{\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d}}$$

$$\widehat{RR} = \frac{P(Enf|Exp)}{P(Enf|No \text{ exp})} \quad IC_{1-\alpha} = e^{\ln \widehat{RR} \pm z_{\alpha/2} \times EE(\ln \widehat{RR})}$$

$$EE(\ln \widehat{RR}) = \sqrt{\frac{1}{Expuestos \text{ enfermos}} - \frac{1}{Expuestos} + \frac{1}{No \text{ expuestos enfermos}} - \frac{1}{No \text{ expuestos}}}$$

Contraste de hipótesis

$$z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}} \quad t = \frac{\bar{x} - \mu_0}{s / \sqrt{n}} \quad t = \frac{\bar{x}_d - \mu_d}{s_d / \sqrt{n}} \quad n = \left(\frac{(|z_\beta| + |z_\alpha|) \times \sigma}{(\mu_1 - \mu_2)} \right)^2$$

$$s_p = \sqrt{\frac{(n_1 - 1) \times s_1^2 + (n_2 - 1) \times s_2^2}{n_1 + n_2 - 2}} \quad t = \frac{\bar{x} - \bar{y}}{s_p \times \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \quad z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

$$\hat{p} = \frac{n_1 \times \widehat{p}_1 + n_2 \times \widehat{p}_2}{n_1 + n_2} = \frac{x_1 + x_2}{n_1 + n_2} \quad z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i}$$

$$\chi^2 = \sum_{i=1}^k \frac{(|O_i - E_i| - 0.5)^2}{E_i}$$

$$r_p = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^N (x_i - \bar{x})^2 \times \sum_{i=1}^N (y_i - \bar{y})^2}} \quad t = \frac{r}{\sqrt{\frac{1-r^2}{n-2}}}$$